MAT 303 Project One Summary Report

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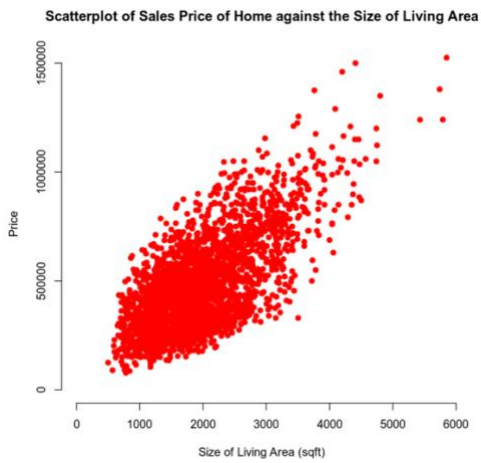
1. Introduction

In this scenario I am working for a real estate company, and I have access to a large set of historical data. The data set includes different attributes of a house like the number of bathrooms and how old the house is. I have been asked to create different regression models to predict sales prices for houses based on different factors. My results can be used to help the company set better prices when listing homes, these better prices can improve the likelihood that listings sell within a reasonable amount of time. The type analyses I will be running is a correlation analysis, exploratory analysis, regression modeling, and predictive analysis.

2. Data Preparation

To be able to identify what variables are considered important it is good to understand what the company wants from me. I know that they want me to create regression models to predict sales price of houses. One important variable here would be price, which is the sales price of the house. Some variables that may affect the price of the house are bedrooms, bathrooms, view, and sqft_living. Depending on how many bathrooms and bedrooms have as well as well as the size of the living area all factor in on how much a home would cost. All these variables will be important to me in predicting sales price of a house. Looking more into the data I see that there are 23 columns and 2692 rows present in the data set, making it a pretty large dataset which is great for accuracy.

3. Model #1 - First Order Regression Model with Quantitative and Qualitative Variables
Correlation Analysis



In the scatterplot of sales price of home against the size of living area I can see a clear positive relationship between the two variables. What this tells me is that as the size of the living area in a house increases, so does the price of said house. Analyzing the scatterplot of sales price of home against the age of the home I can see there really so no trend or relationship between these two variables. This means that on its own age is not really a strong predictor of sales price. Although, I do see a slight negative trend in the plot meaning that older houses might sell for a bit less.

Diving deeper into the relationship between price and these two variables I created a correlation matrix to understand the correlation coefficient between these relationships. The correlation coefficient for sales price of home and size of living area is 0.6895, this is a fairly strong positive correlation and supports my earlier analysis that as size of living area increases so does the price of the home. Viewing

the correlation between price and age I can see that it is a very weak negative correlation having a value of -0.0746, which it is very close to 0. This means that on its own the age of a house has little to no impact on the price of it.

Reporting Results

The general equation of the multiple regression model:

$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6$

The prediction equation of the regression model is:

$\hat{y} = \widehat{\beta_0} + \widehat{\beta_1} x_1 + \widehat{\beta_2} x_2 + \widehat{\beta_3} x_3 + \widehat{\beta_4} x_4 + \widehat{\beta_5} x_5 + \widehat{\beta_6} x_6.$

Where $\hat{y}$ is the predicted value of the sales price of a house, x_1 is the size of living area in sqft, and x_2 is the size of the upper level in sqft, x_3 is the age of the house, and x_4 is the number of bathrooms. x_5 and x_6 are the dummy variables for view.

$\widehat{\beta_0}$, $\widehat{\beta_1}$, $\widehat{\beta_2}$, $\widehat{\beta_3}$, $\widehat{\beta_4}$, and $\widehat{\beta_5}$ are the estimates of β_0, β_1, β_2, β_3, β_4 and β_5.

The equation using my R outputs is:

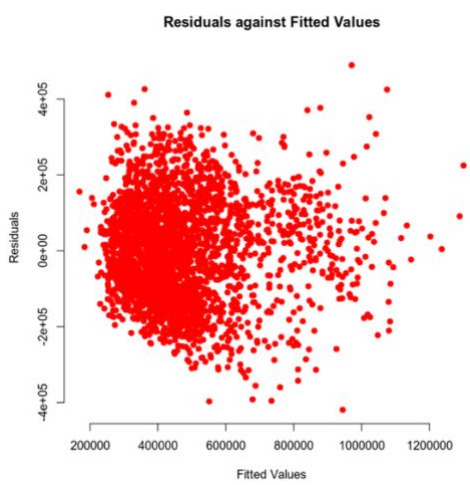
$$\hat{y} = 7.709 \times 10^3 + 1.293 \times 10^2 x_1 + 1.951 \times 10^1 x_2 + 4.397 \times 10^4 x_3 + 1.451 \times 10^3 x_4 + 1.675 \times 10^5 x_5 + 2.490 \times 10^5 x_6$$

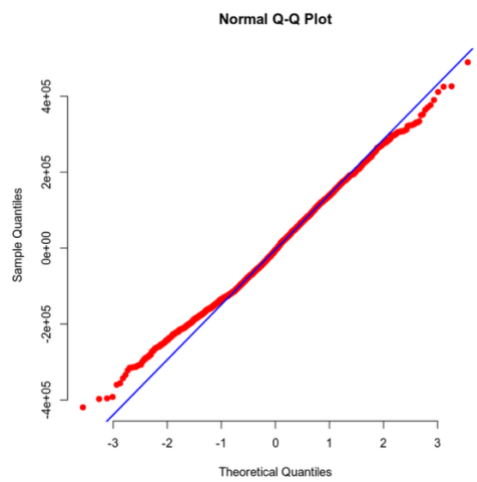
The R-squared value is 0.6029 and the adjusted R-squared value is 0.602. Analyzing the r-squared value first I can say that the model explains about 60.3% of the variation in house sales price. What this tells me is that the predictors used in my model explain just a little over a half of what drives price changes. The explanatory power this model has is somewhat good, I can't say it is perfect but seeing that it is over half it is reasonably good.

The adjusted R-squared value really is no different from the R-squared value with it being 0.602. This means that the R-squared value is adjusted after considering the number of predictors used in the model. Since the value is very close to the R-squared value it suggests that the predictors used are useful and have meaning making the model efficient. I can conclude that the model has solid explanatory power and the predictors chosen are appropriate.

The beta estimate for living area is 1.293×10^2 or 129.3. What this tells me is that while holding other variables constant, for every 1 unit increase in the size of living area(sqft), the price of a home increase by about $129.30. This supports my analysis from earlier as well where the plot showed me that as the size of a home increases so will its price. This goes into detail and tells me that each unit of living space adds about $129 to the home's value.

The beta estimate for lake view is 2.490×10^5 or 249,000. Since my base group is road view, this beta estimate basically tells me how much more the price of a lake view house is compared to that of a house with a road view. So, houses with a lake view are predicted to sell for about $249,000 more than houses with a road view holding all other variables constant.





Analyzing the residuals against fitted values plot I can see that there is no clear curve pattern. The residuals are roughly centered around 0 but the points do not have constant spread. I can see that the spread of the points increases as fitted values increase. Since the residuals do not remain constant throughout the fitted values, I can conclude that the assumption of homoscedasticity is violated based on the evidence shown. As for the normality I can see that the points lie very closely on the reference line in the middle area but not so much in the tails. Specifically in the lower tail, I can see that the points start to deviate away from the reference lines. There is slight deviation in the upper tail, but I believe the most noticeable one is in the lower tail. This could be due to potential outliers, and it may make the distribution of the residuals a bit skewed.

Evaluating Significance of Model

Model	Hypotheses	Test Statistic	P-value	Conclusion
Model #1	$H_0: \beta_1 = \beta_2 \ldots = \beta_k = 0, k = 6$ $H_1: atleast\ one\ \beta_i \neq 0$	679.3 on 6 and 2685 DF	$< 2.2 \times 10^{-16}$	Reject the null hypotheses

Variable	Hypotheses	Test Statistic	P-value	Conclusion
Sqft_living	$H_0: \beta_1 = 0$ $H_1: \beta_1 \neq 0$	15.916	$< 2 \times 10^{-16}$	Reject the null hypotheses
Sqft_above	$H_0: \beta_2 = 0$ $H_1: \beta_2 \neq 0$	2.616	0.00894	Reject the null hypotheses
bathrooms	$H_0: \beta_3 = 0$ $H_1: \beta_3 \neq 0$	7.178	9.13×10^{-13}	Reject the null hypotheses
age	$H_0: \beta_4 = 0$ $H_1: \beta_4 \neq 0$	12.098	$< 2 \times 10^{-16}$	Reject the null hypotheses
view1	$H_0: \beta_5 = 0$ $H_1: \beta_5 \neq 0$	15.640	$< 2 \times 10^{-16}$	Reject the null hypotheses
view2	$H_0: \beta_6 = 0$ $H_1: \beta_6 \neq 0$	20.739	$< 2 \times 10^{-16}$	Reject the null hypotheses

After carrying out the overall F-test and individual beta tests I can conclude that the model and all the variables in the model are statistically significant at the 5% level.

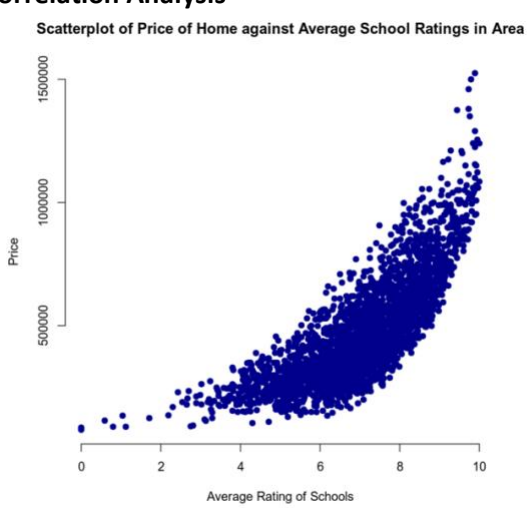
Making Predictions Using Model

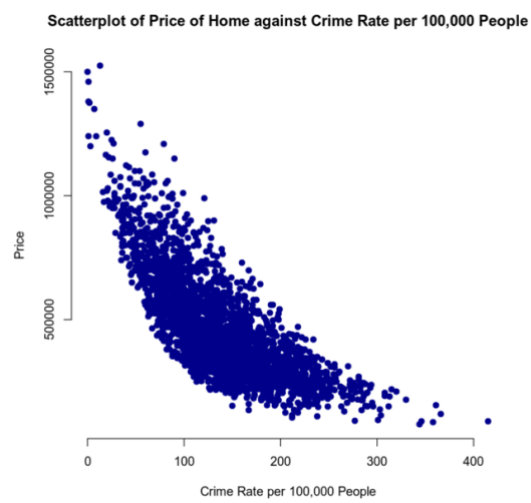
The predicted sales price for a home that has 2150 sqft living area, 1050 sqft upper-level living area, is 15 years old, has 3 bathrooms, and backs out to a road is $459,828.20. The 90% prediction interval for the sales price of this home is (239,563, 680,093.4) and the 90% confidence interval is (446,087.9, 473,568.5). When taking a looking at the prediction interval I am 90% confident that the sales price of a single home will fall between $239,563 and $680,093.4. When looking at the confidence interval I am 90% confident that the true average sales price of all homes will fall between 446,087.9 and 473,568.5. The prediction interval is going to be wider than the confidence interval since the confidence interval only considers the uncertainty in estimating the mean.

The predicted sales price for a home that has 4250 sqft living area, 2100 sqft upper-level living area, is 5 years old, has 5 bathrooms, and backs out to a lake is $1,074,285. The 90% prediction interval for the sales price of this home is (852,522.6, 1,296,048) and the 90% confidence interval is (1,045,117, 1,103,454). When taking a looking at the prediction interval I am 90% confident that the sales price of a single home will fall between $852,522.6 and $1,296,048. When looking at the confidence interval I am 90% confident that the true average sales price of all homes will fall between $1,045,117 and $1,103,454.

4. Model #2 - Complete Second Order Regression Model with Quantitative Variables

Correlation Analysis





A second order model is appropriate in this case mainly because of one reason. When I created these scatterplots I can see a clear pattern. As school ratings increase, price increases, making the relationship between these two variables positive and it creates a curve which requires the use of a second order model. As crime rate increases, price decreases, making the relationship between these variables negative and it also creates a curve making the use of a second model appropriate. Second order models consider the bend or curve in a relationship and nonlinear changes.

Reporting Results

The general equation of the complete second order model:

$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \beta_4 x_1^2 + \beta_5 x_2^2$

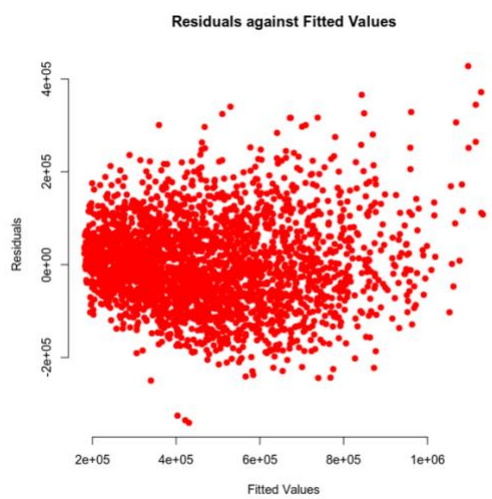
The prediction equation of the second order model is:
$\hat{y} = \widehat{\beta_0} + \widehat{\beta_1}x_1 + \widehat{\beta_2}x_2 + \widehat{\beta_3}x_1x_2 + \widehat{\beta_4}x_1^2 + \widehat{\beta_5}x_2^2.$

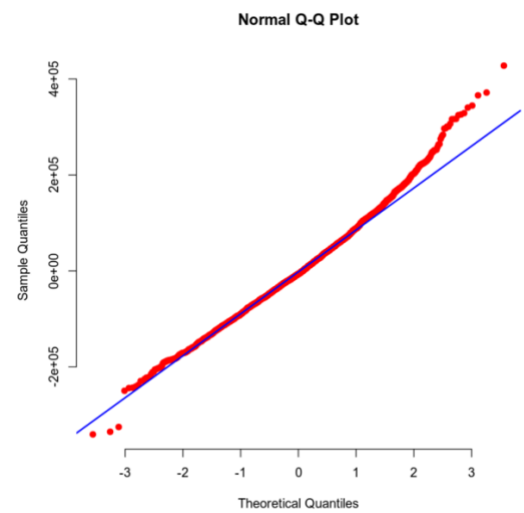
Where $\hat{y}$ is the predicted value of the sales price of a house, x_1 is the average school rating in the area, x_2 crime rate per 100,000 people, and x_1x_2 is the interaction between school rating and crime rate. $\widehat{\beta_0}$, $\widehat{\beta_1}$, $\widehat{\beta_2}$, $\widehat{\beta_3}$, $\widehat{\beta_4}$, and $\widehat{\beta_5}$ are the estimates of β_0, β_1, β_2, β_3, β_4 and β_5.

The equation using my R outputs is:

$$\hat{y} = 7.339 \times 10^5 - 7.375 \times 10^4 x_1 - 3.155 \times 10^3 x_2 - 5.227 \times 10^4 x_1x_2 + 1.165 \times 10^0 x_1^2 + 6.377 \times 10^1 x_2^2$$

The R-squared value is 0.8088 and the adjusted R-squared value is 0.8084. Analyzing the r-squared value first I can say that the model explains about 80.88% of the variation in house sales price. The adjusted R-squared value really is no different from the R-squared value. This means that the R-squared value is adjusted after considering the number of predictors used in the model. Since the value is very close to the R-squared value it suggests that the predictors used are useful and have meaning making the model efficient. I can conclude that the model has strong explanatory power and the predictors chosen are appropriate.





Analyzing the residuals against fitted values plot I can see that there is no clear curve pattern. The residuals are centered around 0 but the points do not have constant spread. I can see that the spread of the points increases as fitted values increase. Since the residuals do not remain constant throughout the fitted values, I can conclude that the assumption of homoscedasticity is violated based on the scatterplot. As for the normality I can see that the points lie very closely on the reference line, apart from the outliers in the lower tail, up until the upper tail when the points start to deviate. There is slight deviation in the lower tail, but I believe the most noticeable one is in the upper tail. This may make the distribution of the residuals right skewed or positively skewed.

Evaluating Significance of Model

Model	Hypotheses	Test Statistic	P-value	Conclusion

Model #2	$H_0: \beta_1 = \beta_2 \ldots = \beta_k = 0, k = 5$ $H_1: atleast\ one\ \beta_i \neq 0$	2272 on 5 and 2686 DF	$< 2.2 \times 10^{-16}$	Reject the null hypotheses

Variable	Hypotheses	Test Statistic	P-value	Conclusion
school_rating	$H_0: \beta_1 = 0$ $H_1: \beta_1 \neq 0$	-3.541	0.000406	Reject the null hypotheses
crime	$H_0: \beta_2 = 0$ $H_1: \beta_2 \neq 0$	-6.07	1.90×10^{-9}	Reject the null hypotheses
school_rating/crime	$H_0: \beta_3 = 0$ $H_1: \beta_3 \neq 0$	-1.077	$< 2 \times 10^{-16}$	Reject the null hypotheses
school_rating squared	$H_0: \beta_4 = 0$ $H_1: \beta_4 \neq 0$	10.497	$< 2 \times 10^{-16}$	Reject the null hypotheses
Crime squared	$H_0: \beta_5 = 0$ $H_1: \beta_5 \neq 0$	8.777	0.281513	Fail to reject the null hypotheses

After carrying out the overall F-test and individual beta tests I can conclude that the model and all the terms in the model except for one, that one being crime squared, are statistically significant at the 5% level.

Making Predictions Using Model

The predicted sales price for a home in an area with average school rating of 9.80 and a crime rate of 81.02 per 100,000 individuals is $874,497. The 90% prediction interval for the sales price of this home is (721,606.2, 1,027,388) and the 90% confidence interval is (863,681.4, 885,312.7). I am 90% confident that the sales price of a single home will fall between $721,606.2 and $1,027,388. I am 90% confident that the true average sales price of all homes will fall between $863,681.4 and $885,312.7.

The predicted sales price for a home in an area with average school rating of 4.28 and a crime rate of 215.50 per 100,000 individuals is $199,706.70. The 90% prediction interval is (46,991.65, 352,421.7) and the 90% confidence interval is (191,753.5, 207,650.9). I am 90% confident that the sales price of a single home will fall between $46,991.65 and $352,421.7. I am 90% confident that the true average sales price of all homes will fall between $1191,753.5 and $207,650.9.

5. Nested Models F-Test

Reporting Results

The general equation of the complete second order model:

$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$

The prediction equation of the second order model is:

$\hat{y} = \widehat{\beta_0} + \widehat{\beta_1} x_1 + \widehat{\beta_2} x_2 + \widehat{\beta_3} x_1 x_2$

The equation using my R outputs is:

$\hat{y} = -410233.37 + 155559.97 x_1 + 2230.07 x_2 - 564.85 x_1 x_2$

Evaluating Significance of Model

Model	Hypotheses	Test Statistic	P-value	Conclusion
Model #3	$H_0: \beta_1 = \beta_2 \ldots = \beta_k = 0, k = 6$ $H_1: atleast\ one\ \beta_i \neq 0$	3573 on 3 and 2688 DF	$< 2.2 \times 10^{-16}$	Reject the null hypotheses

Variable	Hypotheses	Test Statistic	P-value	Conclusion
school_rating	$H_0: \beta_1 = 0$ $H_1: \beta_1 \neq 0$	*49.65*	$< 2 \times 10^{-16}$	Reject the null hypotheses
crime	$H_0: \beta_2 = 0$ $H_1: \beta_2 \neq 0$	*17.20*	$< 2 \times 10^{-16}$	Reject the null hypotheses
school_rating/crime	$H_0: \beta_3 = 0$ $H_1: \beta_3 \neq 0$	*-31.63*	$< 2 \times 10^{-16}$	Reject the null hypotheses

After carrying out the overall F-test and individual beta tests I can conclude that the model and all the variables in the model are statistically significant at the 5% level.

Model Comparison

When comparing two models there is a reduced and a complete model. I know that for models to be considered nested models, one model must have all the terms of the second model and at least one additional term. A complete model would be the model with the additional term while the other model with no additional term is the reduced model.

The general form and prediction equation of the reduced model in this project is:
$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$ and $\hat{y} = \widehat{\beta_0} + \widehat{\beta_1} x_1 + \widehat{\beta_2} x_2 + \widehat{\beta_3} x_1 x_2$

The general form and prediction equation of the complete model in this project is:
$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \beta_4 x_1^2 + \beta_5 x_2^2$
and
$\hat{y} = \widehat{\beta_0} + \widehat{\beta_1} x_1 + \widehat{\beta_2} x_2 + \widehat{\beta_3} x_1 x_2 + \widehat{\beta_4} x_1^2 + \widehat{\beta_5} x_2^2$

Nested Models	Hypotheses	Test Statistic	P-value	Conclusion
Model #2/Model #3	$H_0: \beta_4 = \beta_5 = 0$ $H_1: atleast\ one\ \beta_i \neq 0$	65.20513	2.22716×10^{-28}	Reject the null hypotheses

Since this value is less than 0.05, we can reject the null hypothesis and conclude that at least one of the beta estimates for the squared terms is meaningful and the complete model is necessary.

The results suggest that house prices are strongly influenced by factors such as school ratings and crime rates, and that their effects are not always linear. Higher school ratings are associated with higher house prices, and higher crime rates will of course reduce house prices. Although, the effect of these variables can change depending on how they interact with each other and what level they are at, which is why adding interaction and squared terms improves the model.

Overall, model #2 provides the most accurate and informative prediction of house prices because it explains a larger portion of price variation and accounts for the curved relationships between different

factors and housing value. In simpler terms, this means that house prices are better predicted when we account for both the direct effects and the nonlinear way these factors influence house prices.

The practical importance is that the results show that larger homes, higher school ratings, and better views tend to increase house prices, while higher crime rates tend to decrease them. Because the chosen model explains a large portion of the variation in housing prices, it can be a useful tool for estimating property values, evaluating neighborhoods and schools, and understanding the factors that drive housing market prices.